Nova SBE Econometrics

Classes 6 & 7
Dummy Variables
Instrumental Variables

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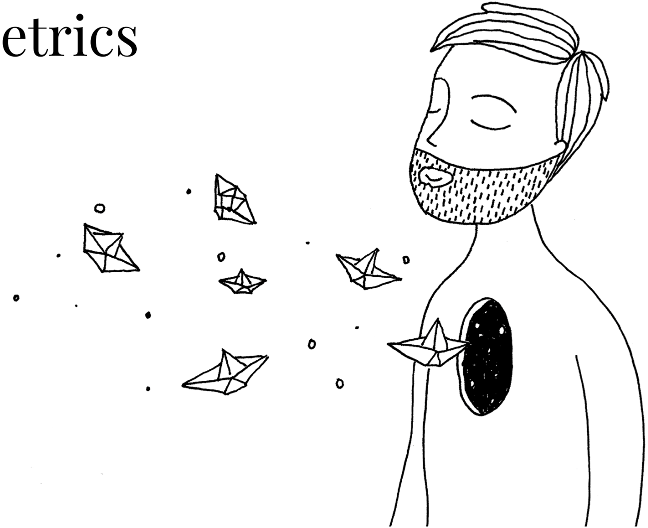

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Nova SBE Econometrics

Dummy Variables & Qualitative Information

Debrief: Why Dummy Variables?

Key idea. Many economic variables are *qualitative*: gender, marital status, industry, region, position played.

We encode them as binary (0/1) **dummy variables** and include them in OLS regressions.

Interpretation rule

The coefficient on a dummy measures the *ceteris-paribus difference in the conditional mean* of y between that group and the **reference (base) category**.

Dummy variable trap. With m mutually exclusive categories, include at most $m-1$ dummies alongside an intercept. Including all m creates *perfect multicollinearity*.

Debrief: Polynomial Terms & Interactions

Quadratic model:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots, \qquad \frac{\partial y}{\partial x} = \beta_1 + 2\beta_2 x.$$

Turning point (optimum): $x^* = -\hat{\beta}_1 \,/\, (2\hat{\beta}_2)$.

Interaction terms:

$$y = \beta_0 + \beta_1 x + \beta_2 D + \beta_3 (D \cdot x) + u, \qquad \frac{\partial y}{\partial x} = \beta_1 + \beta_3 D.$$

F-test for joint significance (q restrictions):

$$F = \frac{(R^2_{\text{ur}} - R^2_{\text{r}})/q}{(1 - R^2_{\text{ur}})/(n - k - 1)} \;\sim\; F(q,\, n - k - 1).$$

Debrief: Individual Significance (t-test)

$$t = \frac{\hat{\beta}_j}{\widehat{\mathrm{se}}(\hat{\beta}_j)} \sim t_{n-k-1}.$$

- Two-tailed ($H_a : \beta_j \neq 0$): reject if $|t| > t_c$.
- One-tailed ($H_a : \beta_j > 0$): reject if $t > t_c$.

Large-n: $t_c \approx 1.96$ (two-tailed, 5%) and $t_c \approx 1.645$ (one-tailed, 5%).

Testing a linear combination $H_0 : \beta_j + \beta_k = 0$

$$t = \frac{\hat{\beta}_j + \hat{\beta}_k}{\sqrt{\widehat{\mathrm{V}}(\hat{\beta}_j) + \widehat{\mathrm{V}}(\hat{\beta}_k) + 2\,\widehat{\mathrm{Cov}}(\hat{\beta}_j, \hat{\beta}_k)}}$$

Exercise 5.5 — Basketball Players: Data

Variables:

- `points` — points scored per game
- `exper` — years as a professional player
- `guard`, `forward` — position dummies (reference: *center*)

Estimated equation ($n = 269$, $R^2 = 0.091$, $\bar{R}^2 = 0.077$):

$$\widehat{\texttt{points}} = \underset{(1.18)}{4.76} + \underset{(0.33)}{1.28}\,\texttt{exper} - \underset{(0.024)}{0.072}\,\texttt{exper}^2 + \underset{(1.00)}{2.31}\,\texttt{guard} + \underset{(1.00)}{1.54}\,\texttt{forward}$$

Exercise 5.5 — Questions

skip to Exercise 5.9 →

(a) Interpret the regression coefficients.
(b) Why can't we include a `center` dummy?
(c) Holding experience fixed, does a guard score more than a center? Is the difference statistically significant?
(d) Using the restricted model below, test whether experience influences points.
(e) Extended model with `marr`: do married players score more points?
(f) Model with marital-status interactions (see table): is marital status jointly significant?
(g) From the interaction model, what is the optimal number of years of experience for married and unmarried players?

Restricted model for (d) ($n = 269$, $R^2 = 0.020$):

$$\widehat{\texttt{points}} = \underset{(0.86)}{8.52} + \underset{(1.03)}{2.40}\,\texttt{guard} + \underset{(1.03)}{1.68}\,\texttt{forward}$$

Exercise 5.5(a) — Interpreting Coefficients

Marginal effect of experience:

$$\frac{\partial\,\widehat{\texttt{points}}}{\partial\,\texttt{exper}} = 1.28 - 2 \times 0.072 \times \texttt{exper} = 1.28 - 0.144\,\texttt{exper}.$$

- Effect is *positive but decreasing*.
- Turning point: $\texttt{exper}^* = 1.28/0.144 \approx 8.9$ years.
- 1st year: $+1.28$ pts; 5th year: $1.28 - 0.144 \times 4 = 0.70$ pts.

Position dummies (reference = center):

- Guards score **2.31 more points** per game than centers, *c.p.*
- Forwards score **1.54 more points** per game than centers, *c.p.*

← back to problem

Exercise 5.5(b) — Dummy Variable Trap

Every player occupies *exactly one* position:

$$\texttt{guard} + \texttt{forward} + \texttt{center} = 1 \quad \forall\, i.$$

Adding `center` creates a perfect linear combination with the intercept column:

$$\texttt{center} = 1 - \texttt{guard} - \texttt{forward}.$$

$\Rightarrow$ **Perfect multicollinearity**: $(X'X)$ is singular and OLS is undefined.

Rule

With m mutually exclusive categories, include at most $m-1$ dummies and an intercept. The omitted category is the **reference group**.

Alternative: drop the intercept and include all m dummies, but group comparisons are less straightforward.

← back to problem

Exercise 5.5(c) — Guards vs. Centers

$H_0 : \beta_3 = 0$ vs. $H_a : \beta_3 > 0$

$$t = \frac{2.31}{1.00} = 2.31 \qquad t_c \approx 1.645 \text{ (one-tailed, 5\%, } df = 264)$$

Since $2.31 > 1.645 \Rightarrow$ **reject** H_0.

Conclusion

Statistically significant evidence that guards score more points per game than centers, holding experience fixed.

← back to problem

Exercise 5.5(d) — Is Experience Significant?

$H_0 : \beta_1 = \beta_2 = 0$ vs. H_a : not H_0

F-test ($q = 2$, $R^2_{\text{ur}} = 0.091$, $R^2_{\text{r}} = 0.020$):

$$F = \frac{(0.091 - 0.020)/2}{(1 - 0.091)/(269 - 4 - 1)} = \frac{0.0355}{0.003443} = 10.31$$

$F_{0.05}(2, 264) \approx 3.00$. Since $10.31 \gg 3.00 \Rightarrow$ **reject** H_0.

Conclusion

Strong evidence that experience is jointly significant in explaining points scored per game.

← back to problem

Exercise 5.5(e) — Marital Status (simple)

Extended model:

$$\widehat{\texttt{points}} = 4.70 + 1.23\,\texttt{exper} - 0.07\,\texttt{exper}^2 + 2.29\,\texttt{guard} + 1.54\,\texttt{forward} + \underset{(0.74)}{0.58}\,\texttt{marr}$$

$H_0 : \beta_5 = 0$ vs. $H_a : \beta_5 > 0$

$$t = \frac{0.58}{0.74} = 0.78 \ < \ 1.645 \ \Rightarrow \ \textbf{fail to reject } H_0.$$

Conclusion

No statistically significant evidence that married players score more points, holding position and experience fixed.

← back to problem

Exercise 5.5(f) — Interaction Model & F-test

	Coef.	Std. Err.	t	$P>\|t\|$
`exper`	0.703	0.434	1.62	0.107
`expersq`	−0.030	0.033	−0.90	0.367
`guard`	2.251	1.000	2.25	0.025
`forward`	1.629	1.002	1.63	0.105
`marr`	−2.538	2.038	−1.24	0.214
`marr_exper`	1.280	0.682	1.88	0.062
`marr_expersq`	−0.094	0.049	−1.92	0.057
Constant	5.816	1.349	4.31	0.000
$n = 269$, $R^2 = 0.1058$, $\bar{R}^2 = 0.0818$, $F(7, 261) = 4.41$				

$H_0 : \beta_5 = \beta_6 = \beta_7 = 0$ vs. H_a : not H_0

$$F = \frac{(0.1058 - 0.091)/3}{(1 - 0.1058)/(269 - 7 - 1)} = 1.44 \; < \; F_{0.05}(3, 261) \approx 2.64 \; \Rightarrow \; \textbf{fail to reject } H_0.$$

← back to problem

Exercise 5.5(g) — Optimal Experience

$$\frac{\partial\,\widehat{\texttt{points}}}{\partial\,\texttt{exper}} = (0.703 - 0.060\,\texttt{exper}) + \texttt{marr}\,(1.280 - 0.188\,\texttt{exper})$$

Unmarried ($\texttt{marr}=0$):

$$0.703 - 0.060\,\texttt{exper}^* = 0 \;\Rightarrow\; \texttt{exper}^* \approx 11.7 \text{ years}$$

Married ($\texttt{marr}=1$):

$$1.983 - 0.248\,\texttt{exper}^* = 0 \;\Rightarrow\; \texttt{exper}^* \approx 8.0 \text{ years}$$

Takeaway

Married players peak earlier (≈ 8 yrs) than unmarried players (≈ 12 yrs) — recall though that marital status is jointly insignificant.

← back to problem

Exercise 5.9 — SAT Scores: Data

Variables: `sat` (combined SAT score), `hsize` (graduating class size, in hundreds), `female`, `black`

Estimated equation ($n = 4{,}137$, $R^2 = 0.0858$):

$$\widehat{\texttt{sat}} = \underset{(6.29)}{1028.10} + \underset{(3.83)}{19.30}\,\texttt{hsize} - \underset{(0.53)}{2.19}\,\texttt{hsize}^2 - \underset{(4.29)}{45.09}\,\texttt{female} - \underset{(12.71)}{169.81}\,\texttt{black} + \underset{(18.15)}{62.31}\,\texttt{female} \times \texttt{black}$$

Reference group: *nonblack males* (`female`=0, `black`=0).

Exercise 5.9 — Questions

skip to Section 2: Instrumental Variables →

(a) Is there strong evidence that $\texttt{hsize}^2$ should be included? What is the optimal high-school size?

(b) Holding `hsize` fixed, what is the estimated difference in SAT scores between nonblack females and nonblack males? How significant is it?

(c) What is the estimated difference between nonblack males and black males? Test whether the difference is zero.

(d) What is the estimated difference between black females and nonblack females? What would you need to test its significance?

Exercise 5.9(a) — Quadratic Term & Optimal Size

$H_0 : \beta_2 = 0$ vs. $H_a : \beta_2 \neq 0$:

$$t = \frac{-2.19}{0.53} = -4.13, \qquad |-4.13| > 1.96 \Rightarrow \textbf{reject } H_0.$$

Optimal class size:

$$19.30 - 2 \times 2.19\,\texttt{hsize}^* = 0 \Rightarrow \texttt{hsize}^* = \frac{19.30}{4.38} = 4.41$$

Interpretation

The optimal graduating class size is $4.41 \times 100 =$ **441** students. Classes smaller or larger are associated with lower average SAT scores.

← back to problem

Exercise 5.9(b) — Nonblack Females vs. Males

$$\mathrm{E}[\texttt{sat} \mid \mathsf{f} = 1, \mathsf{b} = 0] = \hat{\beta}_0 + \hat{\beta}_3$$
$$\mathrm{E}[\texttt{sat} \mid \mathsf{f} = 0, \mathsf{b} = 0] = \hat{\beta}_0$$

Difference: $\hat{\beta}_3 = -45.09$

$H_0 : \beta_3 = 0$ vs. $H_a : \beta_3 \neq 0$:

$$t = \frac{-45.09}{4.29} = -10.51, \qquad |-10.51| \gg 1.96 \;\Rightarrow\; \textbf{reject } H_0.$$

Conclusion

Nonblack females score on average **45.09 points less** than nonblack males; the difference is highly significant.

← back to problem

Exercise 5.9(c) — Black Males vs. Nonblack Males

$$\mathrm{E}[\texttt{sat} \mid \mathsf{f} = 0, \mathsf{b} = 1] = \hat{\beta}_0 + \hat{\beta}_4$$
$$\mathrm{E}[\texttt{sat} \mid \mathsf{f} = 0, \mathsf{b} = 0] = \hat{\beta}_0$$

Difference: $\hat{\beta}_4 = -169.81$

$H_0 : \beta_4 = 0$ vs. $H_a : \beta_4 \neq 0$:

$$t = \frac{-169.81}{12.71} = -13.36, \qquad |-13.36| \gg 1.96 \;\Rightarrow\; \textbf{reject } H_0.$$

Conclusion

Black males score on average **169.81 points less** than nonblack males; the difference is highly significant.

← back to problem

Exercise 5.9(d) — Black Females vs. Nonblack Females

$$\mathrm{E}[\texttt{sat} \mid \mathsf{f} = 1, \mathsf{b} = 1] = \hat{\beta}_0 + \hat{\beta}_3 + \hat{\beta}_4 + \hat{\beta}_5$$
$$\mathrm{E}[\texttt{sat} \mid \mathsf{f} = 1, \mathsf{b} = 0] = \hat{\beta}_0 + \hat{\beta}_3$$

Difference: $\hat{\beta}_4 + \hat{\beta}_5 = -169.81 + 62.31 = -107.50$
$H_0 : \beta_4 + \beta_5 = 0$ vs. H_a : not H_0

$$t = \frac{\hat{\beta}_4 + \hat{\beta}_5}{\sqrt{\widehat{\mathrm{V}}(\hat{\beta}_4) + \widehat{\mathrm{V}}(\hat{\beta}_5) + 2\,\widehat{\mathrm{Cov}}(\hat{\beta}_4, \hat{\beta}_5)}}$$

Alternative

Re-parameterise: let $\theta = \beta_4 + \beta_5$. Re-run with `black` coding the gap between black and nonblack females directly $\Rightarrow$ standard error available from the regression.

← back to problem

Nova SBE Econometrics

Instrumental Variable Estimation

Debrief: The Endogeneity Problem

OLS requires $\mathrm{E}[u \mid X] = 0$. **Endogeneity** arises from:

- **Omitted variable bias** — confounder z affects both X and y.
- **Simultaneity** — y and X are jointly determined (e.g. supply & demand).
- **Measurement error** — X observed with noise.

Consequence

OLS is *biased and inconsistent* when $\mathrm{Cov}(X, u) \neq 0$. The direction of bias depends on the sign of that covariance.

Debrief: Instrumental Variables & 2SLS

Valid instrument Z for endogenous X:

1. **Relevance**: $\mathrm{Cov}(Z, X) \neq 0$ *(testable: first-stage F)*
2. **Exogeneity**: $\mathrm{Cov}(Z, u) = 0$ *(not directly testable with one instrument)*

Two-Stage Least Squares (2SLS):

Stage 1 Regress X on Z (and controls) $\rightarrow$ obtain $\hat{X}$.

Stage 2 Regress y on $\hat{X}$ (and controls).

2SLS is consistent under relevance + exogeneity, but less efficient than OLS when OLS is also consistent.

Debrief: Weak Instruments

Weak instrument: Z explains very little variation in X.
Consequences: 2SLS biased toward OLS; standard errors unreliable; non-normal distribution even in large samples.

Rule of thumb (Stock–Yogo): $F_{\text{first stage}} > 10$.

Bottom line

A strong, exogenous instrument is both *necessary and sufficient* for 2SLS to deliver credible causal estimates.

Exercise 6.1 — Cigarette Demand: Data

Model: $\ln Q_i = \beta_0 + \beta_1 \ln P_i + u_i$

- $\ln Q_i$: log packs per capita sold in state i (1995)
- $\ln P_i$: log after-tax average real price per pack
- 48 continental US states
- **Instrument:** $Z_i = \ln(\texttt{SalesTax}_i)$ — portion of tax from the *general* sales tax

	(1) OLS $\ln Q$	(2) First Stage $\ln P$	(3) 2SLS $\ln Q$
$\ln P$	-1.213^{***} (0.216)		
`salestax`		0.031^{***} (0.005)	
$\widehat{\ln P}$			-1.084^{***} (0.377)
Constant	10.339^{***}	4.617^{***}	9.720^{***}
R^2	0.406	0.471	0.152
F	31.41	40.96	8.28

$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$

Exercise 6.1 — Questions

(a) Is it reasonable to use OLS to estimate the price elasticity of demand? Why?

(b) Discuss the two conditions that $\ln(\texttt{SalesTax})$ must satisfy to be a valid instrument.

(c) How much of the variation in $\ln P$ is explained by the instrument? Relate this to the concept of a weak instrument.

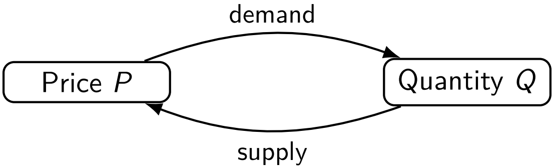
(d) Explain the difference between the second-stage regression (3) and the OLS regression (1).

(e) Interpret the coefficient in (3). Compare with (1) and comment.

(f) What are the main limitations of these models?

Exercise 6.1(a) — Can We Use OLS?

Problem: Simultaneous causality.



Price affects Q demanded, but Q sold also affects equilibrium P. $\Rightarrow \mathrm{Cov}(\ln P,\, u) \neq 0$
$\Rightarrow$ OLS is **biased and inconsistent**.

Solution

Use an instrument that shifts the supply curve (hence price) but is uncorrelated with demand shocks.

← back to problem

Exercise 6.1(b) — Instrument Validity

1. Relevance — $\text{Cov}(Z, \ln P) \neq 0$

The general sales tax is a direct component of the after-tax price per pack: higher tax $\Rightarrow$ higher price.

Testable via first-stage F-statistic.

2. Exogeneity — $\text{Cov}(Z, u) = 0$

The general sales tax applies to all goods and is set by state fiscal policy — not by cigarette-specific demand shocks u_i.

Plausible, but not directly testable with a single instrument.

← back to problem

Exercise 6.1(c) — Weak Instrument Check

From the first-stage regression (column 2):

$$R^2_{\text{first stage}} = 0.471, \qquad F_{\text{first stage}} = 40.96.$$

- The sales tax explains $\approx \mathbf{47}\%$ of the variation in after-tax prices across states.
- $F = 40.96 \gg 10$ — strong evidence *against* a weak instrument.

Reminder

A weak instrument causes 2SLS to be nearly as biased as OLS and renders standard inference unreliable.

← back to problem

Exercise 6.1(d) — Second Stage vs. OLS

Column (1) — OLS: regresses $\ln Q$ on observed $\ln P$, which contains both demand-driven and supply-driven price variation $\Rightarrow$ endogeneity bias.

Column (3) — 2SLS second stage: regresses $\ln Q$ on *fitted* $\widehat{\ln P}$ from Stage 1.

Key insight

$\widehat{\ln P}$ retains only the exogenous (sales-tax-driven) variation in price, purging the component correlated with u. This isolates the causal demand response.

Note: the second-stage R^2 (0.152) is lower than OLS R^2 (0.406) because only part of the price variation is being used.

← back to problem

Exercise 6.1(e) — Interpreting the 2SLS Coefficient

2SLS: $\hat{\beta}_1^{\mathsf{IV}} = -1.084$

Interpretation

A 1% increase in the real price of cigarettes reduces per-capita consumption by $\approx \mathbf{1.08}\%$, *c.p.* Demand is **elastic** ($|\varepsilon| > 1$).

Comparison with OLS ($\hat{\beta}_1^{\mathsf{OLS}} = -1.213$):

- OLS overstates the elasticity (in abs. value) due to simultaneity bias.
- 2SLS reduces the estimate from -1.213 to -1.084: removing endogenous price variation yields a less elastic demand.
- Both suggest fairly elastic demand — somewhat surprising for an addictive good.

← back to problem

Exercise 6.1(f) — Limitations

1. **Omitted variables.** State income affects demand *and* correlates with the sales tax $\Rightarrow$ remaining omitted-variable bias.
2. **Single instrument.** Exogeneity not testable with one instrument; a second would allow an over-identification (Sargan/Hansen J) test.
3. **Cross-section only (1995).** Panel data with state fixed effects would absorb time-invariant confounders.
4. **Homogeneous elasticity.** A single β_1 for all states may miss variation by income, demographics, or culture.

Next step

A **multiple IV** approach with income controls and panel data would substantially improve credibility.

← back to problem